Lecture 9: Bagging and Random Forests

1 Motivation for Bagging: Instability and Aggregation

1.1 Recap: Decision Tree Predictors

Decision trees are nonparametric estimators that partition the feature space $\mathcal{X} \subset \mathbb{R}^p$ into disjoint regions $\{R_1, \ldots, R_K\}$ and produce piecewise-constant predictions. For regression,

$$T(x) = \sum_{k=1}^{K} \hat{c}_k \mathbf{1}\{x \in R_k\}, \qquad \hat{c}_k = \frac{1}{|R_k|} \sum_{i: x_i \in R_k} y_i.$$

For classification, the prediction is typically the majority class in each region.

Trees are attractive because they

- capture nonlinear decision boundaries,
- automatically model interactions,
- require minimal preprocessing,
- are invariant to monotone transformations of predictors.

However, these advantages come at the cost of statistical instability.

1.2 Statistical Instability

Let $\mathcal{D}_n = \{(x_i, y_i)\}_{i=1}^n$ denote the training data and let $\hat{f}(x; \mathcal{D}_n)$ be the fitted predictor. A learning procedure is *unstable* if small perturbations of the training data lead to large changes in the fitted function. A natural measure is

$$\mathrm{Var}_{\mathcal{D}_n}\big(\hat{f}(x; \mathcal{D}_n)\big).$$

Deep decision trees are particularly unstable because

- early split decisions propagate globally,
- recursive partitioning is discrete and non-smooth,
- small data changes can alter the tree topology.

Empirically, trees often exhibit low bias but high variance.

1.3 Bias–Variance Motivation

For regression with squared loss,

$$\mathbb{E}(Y - \hat{f}(x))^2 = \text{Bias}^2 + \text{Var} + \sigma^2.$$

Flexible learners such as deep trees and neural networks often operate in a variance-dominated regime. Reducing variance without substantially increasing bias can therefore improve predictive performance, motivating aggregation.

1.4 Model Averaging Principle

Suppose we had access to B independent training datasets $\mathcal{D}_n^{(1)}, \ldots, \mathcal{D}_n^{(B)}$ and fit

$$\hat{f}_b(x) = \hat{f}(x; \mathcal{D}_n^{(b)}).$$

The averaged predictor is

$$\hat{f}_{\text{avg}}(x) = \frac{1}{B} \sum_{b=1}^{B} \hat{f}_b(x).$$

If base learners share the same expectation, averaging preserves bias while reducing variance. Under independence,

$$\text{Var}\big(\hat{f}_{\text{avg}}(x)\big) = \frac{\sigma_f^2(x)}{B}.$$

With pairwise correlation $\rho(x)$,

$$\text{Var}\big(\hat{f}_{\text{avg}}(x)\big) = \rho(x)\sigma_f^2(x) + \frac{1 - \rho(x)}{B}\sigma_f^2(x).$$

Thus ensemble performance depends on

- the number of learners B,
- the correlation among learners.

1.5 From Ideal Averaging to Bagging

Because multiple independent datasets are unavailable in practice, bagging approximates model averaging via bootstrap resampling.

2 Bootstrap Approximation for Bagging

2.1 Empirical Distribution

Let

$$\mathcal{D}_n = \{(x_i, y_i)\}_{i=1}^n$$

be i.i.d. from an unknown distribution P. The bootstrap replaces P with

$$\hat{P}_n = \frac{1}{n}\sum_{i=1}^{n} \delta_{(x_i,y_i)}.$$

A bootstrap sample $\mathcal{D}_n^*$ is obtained by sampling with replacement from $\hat{P}_n$.

2.2 Multinomial Representation

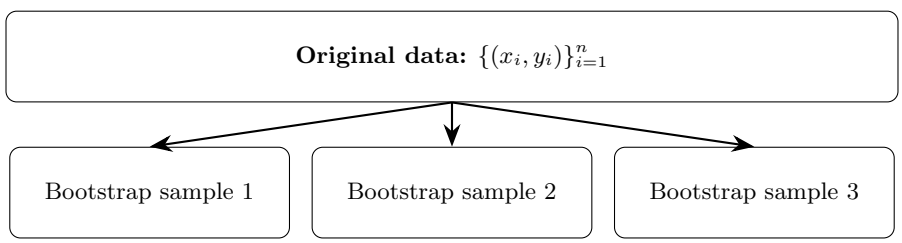
Let N_i^* be the number of times observation i appears. Then

$$(N_1^*, \ldots, N_n^*) \sim \text{Multinomial}\left(n; \frac{1}{n}, \ldots, \frac{1}{n}\right), \qquad \sum_{i=1}^{n} N_i^* = n.$$

2.3 Out-of-Bag Probability

$$\Pr(N_i^* = 0) = \left(1 - \frac{1}{n}\right)^n \to e^{-1} \approx 0.368.$$

Implication. Approximately 36.8% of observations are out-of-bag (OOB) for each tree, enabling internal error estimation without a separate validation set.



Sampling with replacement; some points left out (OOB)

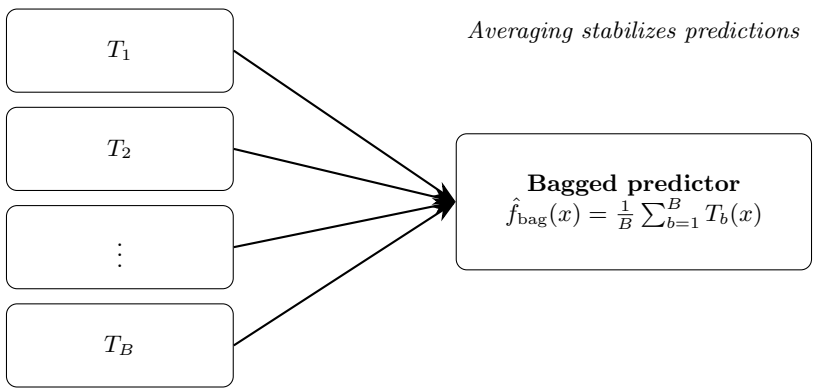
2.4 Bagging

Bagging (Bootstrap Aggregating) proceeds as follows:

1. For $b = 1, \ldots, B$, draw a bootstrap sample.
2. Fit a tree T_b to the bootstrap data.
3. Aggregate predictions:

$$\hat{f}_{\text{bag}}(x) = \frac{1}{B}\sum_{b=1}^{B} T_b(x).$$

Remark. Although decision trees are the most common base learner in bagging, the procedure applies to any unstable learning algorithm. The primary goal of bagging is variance reduction through model averaging.



3 Variance of Bagging: Full Derivation

Fix a point x and consider B base learners

$$\hat{f}_1(x), \ldots, \hat{f}_B(x), \qquad \hat{f}_{\text{avg}}(x) = \frac{1}{B}\sum_{b=1}^{B} \hat{f}_b(x).$$

For notational simplicity, write $Z_b = \hat{f}_b(x)$ and $\bar{Z} = \hat{f}_{\text{avg}}(x)$.

3.1 General covariance expansion

$$\operatorname{Var}(\bar{Z}) = \operatorname{Var}\left(\frac{1}{B}\sum_{b=1}^{B} Z_b\right) = \frac{1}{B^2}\operatorname{Var}\left(\sum_{b=1}^{B} Z_b\right).$$

Expand

$$\operatorname{Var}\left(\sum_{b=1}^{B} Z_b\right) = \sum_{b=1}^{B}\operatorname{Var}(Z_b) + 2\sum_{1\le b<b'\le B}\operatorname{Cov}(Z_b, Z_{b'}).$$

Therefore,

$$\operatorname{Var}(\bar{Z}) = \frac{1}{B^2}\left[\sum_{b=1}^{B}\operatorname{Var}(Z_b) + 2\sum_{1\le b<b'\le B}\operatorname{Cov}(Z_b, Z_{b'})\right]. \tag{1}$$

3.2 Exchangeable case (equal variance and common correlation)

Assume exchangeability at x:

$$\mathrm{Var}(Z_b) = \sigma^2(x) \text{ for all } b, \qquad \mathrm{Corr}(Z_b, Z_{b'}) = \rho(x) \text{ for } b \neq b'.$$

Then $\mathrm{Cov}(Z_b, Z_{b'}) = \rho(x)\sigma^2(x)$ for $b \neq b'$. Plugging into (1),

$$\sum_{b=1}^{B} \mathrm{Var}(Z_b) = B\sigma^2(x), \qquad 2 \sum_{1 \le b < b' \le B} \mathrm{Cov}(Z_b, Z_{b'}) = B(B-1)\rho(x)\sigma^2(x).$$

Hence

$$\mathrm{Var}(\bar{Z}) = \frac{1}{B^2}\left[B\sigma^2(x) + B(B-1)\rho(x)\sigma^2(x)\right] = \frac{\sigma^2(x)}{B} + \frac{B-1}{B}\rho(x)\sigma^2(x),$$

or equivalently

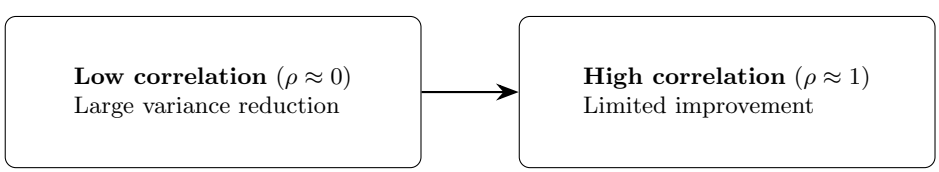
$$\mathrm{Var}\big(\hat{f}_{\mathrm{avg}}(x)\big) = \rho(x)\sigma^2(x) + \frac{1-\rho(x)}{B}\sigma^2(x). \tag{2}$$

3.3 Interpretation

Equation (2) decomposes ensemble variance into:

- a **variance floor** $\rho(x)\sigma^2(x)$ (persists as $B \to \infty$),
- a **vanishing term** $\frac{1-\rho(x)}{B}\sigma^2(x)$ (decays as $1/B$).

Key takeaway. Ensemble performance improves by increasing B, and often more importantly by reducing correlation among ensemble members. This correlation-induced variance floor motivates methods that explicitly decorrelate the ensemble members, leading to random forests.



Reducing correlation is crucial for bagging performance

4 Random Forests: Reducing Correlation in Bagging

4.1 Limitation of Bagging

The variance decomposition

$$\mathrm{Var}\big(\hat{f}_{\mathrm{bag}}(x)\big) = \rho(x)\sigma^2(x) + \frac{1-\rho(x)}{B}\sigma^2(x)$$

shows that averaging alone cannot remove the correlation-driven variance floor $\rho(x)\sigma^2(x)$. In practice, trees grown on bootstrap samples can be strongly correlated because they repeatedly select the same dominant predictors near the top splits.

4.2 Key Idea

Random forests decorrelate trees by modifying how splits are chosen:

At each split, consider only a random subset of predictors.

4.3 Algorithm

Let m denote the number of candidate predictors considered at each split, typically $m < p$. For $b = 1, \ldots, B$:

1. Draw a bootstrap sample from the training data.
2. Grow an unpruned decision tree.
3. At each node:
   - randomly select m predictors from $\{1, \ldots, p\}$,
   - choose the best split using only these m predictors.

The final predictor is

$$\hat{f}_{\mathrm{RF}}(x) = \frac{1}{B}\sum_{b=1}^{B} T_b^{\mathrm{RF}}(x).$$

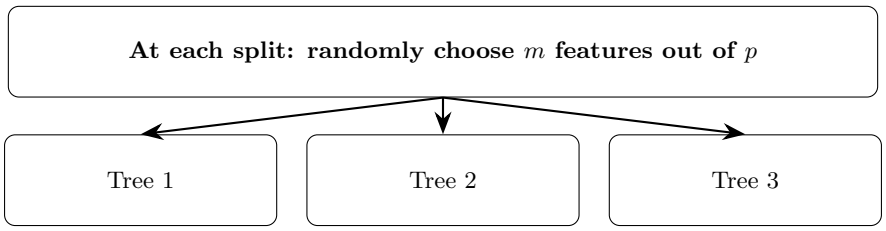
4.4 Choice of m

The parameter m controls the strength–correlation trade-off:

- smaller $m \Rightarrow$ lower correlation but weaker trees,
- larger $m \Rightarrow$ stronger trees but higher correlation.

Common defaults are

$$m = \sqrt{p} \ \text{ (classification)}, \qquad m = \frac{p}{3} \ \text{ (regression)}.$$



Different features ⇒ lower correlation ⇒ lower variance

5 Out-of-Bag (OOB) Error Estimation

A key advantage of bagging and random forests is that they provide an internal estimate of prediction error without requiring a separate validation set.

5.1 OOB Predictions

For observation i, define the index set of trees for which i is out-of-bag:

$$\mathcal{B}_i = \{b : (x_i, y_i) \text{ not used in training tree } b\}.$$

The OOB prediction at x_i averages only over trees that did not see i:

$$\hat{f}_{\text{OOB}}(x_i) = \frac{1}{|\mathcal{B}_i|} \sum_{b \in \mathcal{B}_i} T_b(x_i).$$

5.2 OOB Error Formula

Regression (OOB MSE).

$$\text{Err}_{\text{OOB}} = \frac{1}{n} \sum_{i=1}^{n} \left(y_i - \hat{f}_{\text{OOB}}(x_i) \right)^2.$$

Classification (OOB Misclassification Rate). Let $\hat{y}_i^{\text{OOB}}$ be the majority vote among $\{T_b(x_i) : b \in \mathcal{B}_i\}$. Then

$$\text{Err}_{\text{OOB}} = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\{\hat{y}_i^{\text{OOB}} \neq y_i\}.$$

Why it works. Each OOB prediction for x_i is made using trees trained without observation i, so Err_{OOB} is approximately unbiased for the test error.

6 Variable Importance in Random Forests

Random forests provide built-in measures of predictor importance. Two common importance measures are mean decrease in impurity and permutation importance.

6.1 Mean Decrease in Impurity (MDI)

Each split reduces node impurity. For regression, impurity is typically variance; for classification, the Gini index or entropy.

For a split using variable j, define the impurity decrease

$$\Delta I = I(\text{parent}) - \left(\frac{n_L}{n} I(\text{left}) + \frac{n_R}{n} I(\text{right})\right).$$

The importance of variable j sums ΔI over all splits using j, averaged across trees:

$$\mathrm{VI}_{\mathrm{MDI}}(j) = \frac{1}{B} \sum_{b=1}^{B} \sum_{\substack{\text{splits in } T_b \\ \text{using } j}} \Delta I.$$

MDI is fast to compute but can be biased toward predictors with many split points (e.g., continuous variables).

6.2 Permutation Importance (Mean Decrease in Accuracy)

Permutation importance measures the increase in prediction error when a variable is permuted. For each tree b:

1. Compute OOB error Err_b on OOB data.
2. Permute variable j in the OOB data and recompute error $\mathrm{Err}_{b,j}^{\mathrm{perm}}$.

Define

$$\mathrm{VI}_{\mathrm{perm}}(j) = \frac{1}{B} \sum_{b=1}^{B} \left(\mathrm{Err}_{b,j}^{\mathrm{perm}} - \mathrm{Err}_b\right).$$

Permutation importance is more model-agnostic and typically less biased than MDI, but can be affected by correlated predictors.

7 Summary

- Bagging reduces variance via averaging across bootstrap-trained models.
- OOB predictions provide an internal estimate of test error.
- Correlation limits bagging performance.

- Random forests reduce correlation via feature subsampling.
- Random forests provide variable importance measures.